Impact of Spring Coefficient Variations on the Acoustic Properties of Piano Strings

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Abstract

This research experiment explores the impact of varying the spring coefficient of a piano string on the acoustic properties of its produced sound waves when struck. This study examines the relationship between a piano string's stiffness or spring coefficient and the resulting wavelength of its sound waves. By investigating the influence a piano string's stiffness has on its sound, this research paper aims to provide practical knowledge for piano engineers and those explicitly working in string tuning. By providing insight into how a piano string's spring coefficient can affect its resulting pitch or sound, this study aims to improve our understanding of sound wave analysis within piano strings and enhance the current approach to developing and maintaining pianos. The primary research method was a programmed simulation model that implemented fundamental physics principles and external data to simulate the oscillations of a piano string. Our simulation model integrated external online repositories and sources to define and simulate the realistic properties of a piano string. This study's results indicate an inverse correlation between a piano string's stiffness and the wavelength of its produced sound wave. Our model's data suggest that as the piano string's stiffness increases, the frequency it produces increases, and thus, the wavelength of its sound wave decreases. Ultimately, our research highlights that a piano string's spring coefficient or stiffness plays a significant role in shaping the acoustic characteristics of its sound.

Index Terms— String stiffness, Spring coefficient, Piano acoustics, Soundwave analysis

1 Introduction

The piano is a captivating instrument that can produce a vast range of sounds, from delicate to powerful notes. But how exactly does a piano produce its sound? Due to the delicate interplay between a piano's strings and the hammers within the instrument's insides, various pitches and frequencies can be made. Once a piano key is pressed, a hammer within the inside of a piano strikes a group of strings; these strings are typically in groups of three. After the hammer strikes down, the tension within the strings causes them to vibrate in a periodic motion that moves back and forth [1]. The vibration of each piano string then generates a mechanical sound wave that travels through the surrounding air and atmosphere. Given that a standard grand piano has 88 keys, each capable of producing a unique sound and pitch, it is crucial to comprehend the factors that can influence the sound waves of each piano string [1]. This is because even the most minor changes in a piano string's physical properties, such as the tension it experiences, can significantly impact the sounds it produces [2].

The acoustic properties of a piano string's sound waves are determined by the characteristics of the piano string itself. One of these physical properties of the string is the spring coefficient, which refers to how stiff the spring within the piano string is. By understanding the impact of a piano spring's stiffness on the acoustics of the sound waves produced, piano engineers can understand the significance of the intrinsic properties of a string on its resulting sound. Additionally, the oscillations of a piano string can be deduced by implementing physics principles. Hooke's Law, which dives into the restoring force that a piano string experiences once it is displaced or struck, provides a fundamental relationship between the tension and stiffness of the string [2]. Furthermore, incorporating simple harmonic motion equations, which describe the periodic motion of a piano string, can assist with the visualization of a string's oscillations. By employing these laws and principles into a theoretical framework, a relationship between a piano string's stiffness or spring coefficient and the wavelength of its produced sound wave can be determined.

This research paper aims to answer the question: *How does varying the spring coefficient of a piano string influence the sound produced when a piano key is struck?* By exploring this field of acoustic physics, we can provide valuable knowledge to those who work in piano engineering or tuning on the significance of a piano string's stiffness. This knowledge aims to improve

the current approach to developing and maintaining standard pianos. Additionally, the primary research method in this study was a programmed simulation model. This paper dives into the question above by developing a model based on fundamental physics principles and external data sources on the properties of a piano string. This paper presents and analyzes the results of our simulation model to discover the relationship between a piano string's stiffness and the wavelength it can produce.

2 Literature Review

The physics behind sound production in piano strings has been extensively researched. Multiple studies examine the mechanical aspects of a piano string's sound waves. Some studies have also dived into the impact of a piano string's physical properties on the sound it produces; e.g., studies discuss the effects of the tension exerted on a piano string. Overall, those who focus on piano engineering and tuning must have a strong understanding of these concepts, as it can help them better understand how varying string properties can influence the frequency and sound it produces.

2.1 Basis of Sound Production

The way that a piano string can produce sound arises from its ability to vibrate and oscillate back and forth. When a piano key is struck, a hammer within the piano's insides strikes down on the strikes [3]. The strike of the hammer onto the strings and the tension within the strings themselves allow for the string's vibration. This vibration then generates a mechanical wave; this wave travels through the inner dynamics of the piano string along its length, thus generating sound [3]. The mechanical wave generated can vary depending on the string's characteristics. Depending on the sound wave's frequency or how fast the string oscillates back and forth, it can allow for the varying pitches that humans perceive [3].

2.2 Impact of External Forces

The tension a piano string undergoes or experiences is directly proportional to its frequency of oscillation. Specifically, increasing the piano string's tension results in a higher frequency and, thus, an increased perceived pitch [3].

This phenomenon is accompanied by a correlation between the soundwave's frequency and wavelength. As shown by the formula $v = \lambda f$, which v represents the velocity of the sound wave, λ the wavelength, and f the oscillation frequency, the wavelength λ of a piano string's sound wave is indirectly proportional to its oscillation frequency f. This theory infers that when a piano string's sound wave frequency increases, its wavelength decreases. By understanding the relationship between string tension, frequency, and wavelength, we can better understand how to manipulate these different factors to produce varying sounds in a piano.

Moreover, the intensity of a strike on a piano key can change the timbre of its sound. Timbre refers to the quality or color of the sound. Hence, the more intense a strike is on a piano key, the more timbre it is, and the 'brighter' the sound produced [4]. The different variations of hammer sizes and mass in the backend of the piano also impact the timbre of a piano string's sound. As the hammer's size and mass increase, the sound's timbre or intensity increases correspondingly [5]. In addition, the hammer's contact time with the string can influence the sound produced [4]; the contact time of a string refers to how fast the key is struck. This theory means that a decrease in the hammer contact time corresponds to a more impulsive strike and, thus, a higher frequency.

2.3 Composition of Mechanical Wave

Diving deeper into the mechanics of a piano string's sound waves, studies have shown that the wave generated by a string's vibration consists of longitudinal and transverse waves [3]. Some studies indicate that these sound waves are predominantly transverse waves, where according to Hooke's Law, $F = kx$, (where F represents the force experienced by a spring, k the stiffness or spring coefficient, and x the displacement of the string from equilibrium), the direction of the wave is perpendicular to the displacement of the string [2].

The presumed predominance of transverse waves does not necessarily overshadow the presence of longitudinal waves. Studies indicate that models used to calculate potential energy density in string waves often exclude the longitudinal motion of the string. This misinterpretation is said to have led to inconsistencies in the formulas used to calculate the potential energy density [6]. Hence, incorporating the longitudinal motion within a piano string's soundwave could lead to a more accurate representation of the energy distribution within vibrating piano strings. Understanding the significance of

both longitudinal and transverse motion within sound waves makes it easier to visualize how the complexity of wave interactions can contribute to a piano's overall sound. Past research studies have also discussed the impact of the medium through which the sound waves of a piano string travel. Studies show that a medium with a greater density results in a slower speed of the soundwave [3].

2.4 Impact of Physical Components

The piano is an instrument with many physical characteristics and can produce various sounds with different timbres and lengths. One way that a piano manages to amplify its sound is through its physical component called the soundboard. Within a conventional grand piano, the soundboard is a large, thin wooden plate that goes over the top of the piano. How the soundboard functions is that it helps to displace more air than the sound waves by the strings alone, leading to an intensified and amplified version of the piano strings [5]. Moreover, the soundboards of a standard piano are reinforced with ribs to maintain its stiffness and produce a stronger and more amplified sound. This aspect of rigidity supports the soundboard's ability to vibrate in response to the string's vibrations, improving sound amplification [7].

Another physical property of a piano string that impacts its production of sound is the strings' materials. The majority of modern piano strings are made of high-carbon steel and copper, both of which are capable of maintaining high tensions, allowing for repetitive striking without much deterioration [8]. Each string across the broad spectrum of piano frequencies produces a unique sound due to its length and thickness. Previous studies demonstrate that the thickness of a string is directly proportional to the wavelength of the string [5]. This theory explains why the lower frequencies of a piano are generated through thicker strings and higher frequencies are generated by the thinner strings.

Overall, existing research has provided a strong foundation for our current understanding of the mechanics and acoustics behind the sound of a piano string. However, there needs to be more specific information regarding the effects of a piano string's intricate properties on the soundwave it produces. More specifically, studies lack detailed information on the impact of a piano string's stiffness on the acoustic output. By implementing current knowledge, this research paper aims to dive deeper into the effect of varying the spring coefficient of a piano string on its resulting wavelength.

3 Methodology

This study is an experimental investigation incorporating simulation models to explore the impact of spring constant variations on the wavelength of a piano string's sound waves. Our simulation model integrates theoretical physics principles and external data to determine how a piano string's stiffness variations can affect its sound production.

3.1 Theoretical Framework and Derivation

Our theoretical framework began with a derivation involving classical and Newtonian mechanics. This derivation aimed to model an equation that describes the relationship between a piano string's stiffness (k) and the wavelength (λ) of its soundwave. This derivation began with Hooke's Law of Elasticity, which states that $F = -kx$, where F represents the force experienced by a spring, k the stiffness or spring coefficient, and x the spring's displacement from equilibrium. By implementing Hooke's Law, we could express the stiffness of a piano string with respect to the tension and displacement that it experienced. This principle was combined with Newton's Second Law of Motion, which states that $F = ma$, where m represents the mass of the piano string and a the acceleration of the string. By implementing Newton's Second Law, we could model the dynamic motion of the piano string with respect to the force it experiences. By merging both of these laws, a relationship between the displacement of a piano string and its resulting acceleration was established with the equation: $-kx = ma$.

We expanded our derivation further by implementing the vibrations of the piano string, similar to the oscillations of a standard spring. The periodic motion of the piano string was modeled through the SHM equation, $x(t) = A\sin(\omega t)$, where A represents the amplitude of oscillation, and ω the angular frequency [2]. Through incorporating this equation, our derivation becomes:

$$-kA\sin(\omega t) = ma$$

Moreover, by applying the rules of calculus and recognizing that the second derivative of displacement is acceleration [9], our equation was rewritten as:

$$-kA\sin(\omega t) = m\frac{d^2x}{dt^2}$$

The equation for the angular frequency ω of a periodic spring-mass system is $\omega = \frac{2\pi}{t}$, where t represents the time it takes for one complete oscillation. This equation can be rewritten as $\omega = 2\pi f$, as $f = \frac{1}{t}$, where f represents the oscillation frequency [10]. Additionally, the equation $v = \lambda f$ describes the relationship between the soundwave's velocity and its wavelength and frequency; v represents the velocity of the soundwave, λ the wavelength, and f the frequency of oscillation. By substituting these relationships, our equation becomes:

$$-kA\sin(\frac{2\pi vt}{\lambda}) = m\frac{d^2x}{dt^2}$$

Through differentiating the SHM equation for the periodic displacement of the piano string, our equation results as:

$$-kA\sin(\frac{2\pi vt}{\lambda}) = \frac{-4mA\pi^2v^2\sin(\frac{2\pi vt}{\lambda})}{\lambda^2}$$

$$k = \frac{4m\pi^2v^2}{\lambda^2}$$

Through the formula, $m = \rho V$, we obtain that the piano string's mass m can be substituted as the product of the string's density and volume, where ρ represents the density of the string and V the volume. Since the shape of a piano string is cylindrical [11], the volume of a piano string can be derived as: $V = \pi r^2 l$, where r represents the radius of the string, and l the length. Thus, implementing this formula gives our final equation:

$$\lambda = \sqrt{\frac{4\rho\pi^3r^2lv^2}{k}}$$

3.2 Implementation of Derivation into Simulation

After deriving the relationship between the piano string's stiffness and its wavelength, a simulation model was designed and programmed in Python to implement this theoretical framework. Throughout the simulation process, three models were developed, each progressively more representative of a piano string, by incorporating data from external sources regarding piano string oscillations. Furthermore, each simulation model utilized Python libraries to assist with data analysis and collection. Specifically, the NumPy library was used to conduct numerical computations and handle large arrays,

while the Matplotlib library helped visualize the data and plot graphs with best-fit lines.

Our initial model implemented our theoretical framework and tested an arbitrary range of spring constant values. By setting the other variables of our equation (the length, mass, radius of the string, and the velocity of the wave) to constant values, this model explored the fundamental relationship between a piano string's stiffness and its resulting wavelength. According to current studies, a piano string's radius varies between 0.0035m to 0.000395m with larger radii on the left end and smaller radii on the right end [12]; hence, a radius of 0.001m was used in this model, an appropriate value within the center of the range. The mass of the string was set to 0.01kg, and the length of the string to 1m. These values represent a piano string's simplified yet realistic properties. Our model generated 50 arbitrary values for the spring coefficients from 1 to 100. By implementing these arbitrary data points, our initial model allowed us to visualize how the string stiffness of a piano string could influence its produced sound waves.

3.3 Refined Model using External Data

Our initial model represented an unrealistic range of spring constants, as it implemented arbitrary values for the stiffness of a string. In our second model, we refined this by employing previously tested and researched data ranges. We implemented the results of Carl Lemer and Michael Yang's paper titled *"Measuring the speed of sound through piano strings and its dependence on tension using longitudinal waves."* This paper provided valuable information on the soundwave analysis of a piano string; specifically, it offered a graph discussing the impact of tension exerted on a piano string on the velocity of its produced soundwave.

The graph above served as a foundation for our refined model as we implemented the ranges of wavelengths and spring constant values from the graph. To develop these two parameters, arbitrary but appropriate values for the soundwave frequency and the piano string's displacement were created. The displacement of the piano string was set to 0.01m, and its oscillation frequency was 500 Hz, both of which fall within the typical range for piano strings [13]. By creating these arbitrary variables, we calculated a more realistic range of wavelengths and spring constants of a piano string. For instance, the lower end of the wavelength range was 8m, which was calculated through the formula $\lambda = v/f$, where the velocity of the wave v was 4000

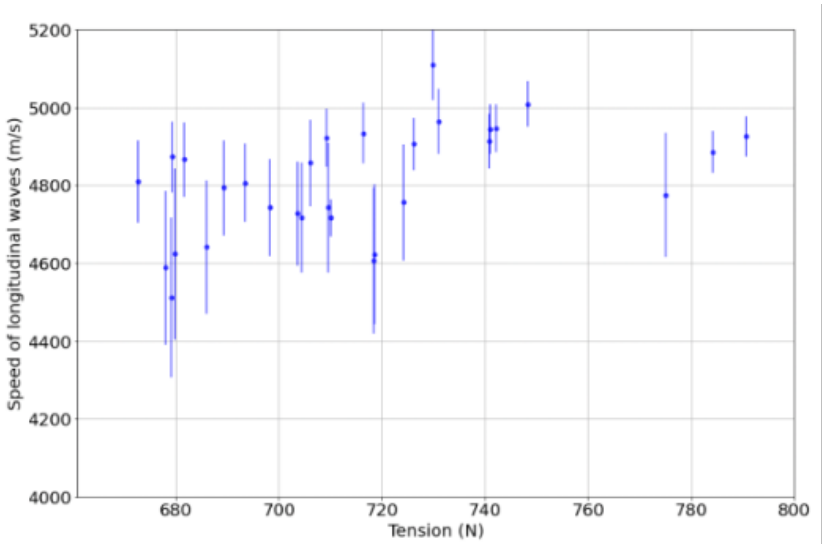


Figure 1: Speed of Longitudinal Wave vs. Tension of Piano String; "Measuring the Speed of Sound through Piano Strings and Its Dependence on Tension Using Longitudinal Waves" by Carl Lemer and Michael Yang

m/s (lower end of the speed of longitudinal wave taken from graph), and the frequency of oscillation f was 500 Hz. After finalizing the ranges for both the wavelengths and the spring constants, our second model generated 30 values for the wavelength and the spring constants, similar to the total amount of data points within the graph. By implementing this external data, our model could more realistically simulate the oscillations of a piano string, providing higher accuracy and precision.

3.4 Final Model: Theoretical vs. Experimental Data

For our simulation model's third and final revision, we closely aligned our model's parameters with the data in the researched graph. By continuing to incorporate the previous ranges of wavelengths and spring constants, we directly input each data point from the graph into the simulation model. This allowed for a more accurate representation of the piano string's vibrations. Furthermore, our final model compared our model's predicted values of spring coefficients to the results of the researched graph. This was achieved by using a computer vision-assisted software called WebPlotDigitzer.

The range of the spring coefficient values was increased to match the range of stiffness values achieved from our second model; this was 66,000 to 80,000 N/m, an accurate measurement of the spring constant of a piano string [14]. Additionally, our final model implemented a velocity value of 4100 m/s, an arbitrary value that ensured the raw data from the paper would match the range of our model's values. This value was decided to create an external coefficient for the velocity, which allowed our model to account for external factors that could have influenced the raw data values.

4 Results

4.1 Initial Model: Arbitrary Spring Constant Values

The two graphs below demonstrate the results of our initial simulation model, where randomized values for the spring constant were generated; this range was 1 to 100 N/m. Our initial model aimed to visualize the basic relationship between a piano string's stiffness and its resulting wavelength. Figure 2 below indicates an indirect relationship between the spring constant of a piano string and the wavelength of its produced soundwave. The negative slope of the best-fit line confirms this trend, as when the spring constant approaches 100 N/m, the wavelength of the sound wave decreases exponentially. In contrast, Figure 3 below depicts a positive correlation between the frequency of a piano string's sound wave and its spring constant. The slope of the best-fit line indicates that as the spring constant increases, the frequency likewise increases; this reveals that higher stiffnesses of piano strings lead to increased frequencies and pitches. This is consistent with the formula $v = \lambda f$, as wavelength and frequency have an indirect relationship.

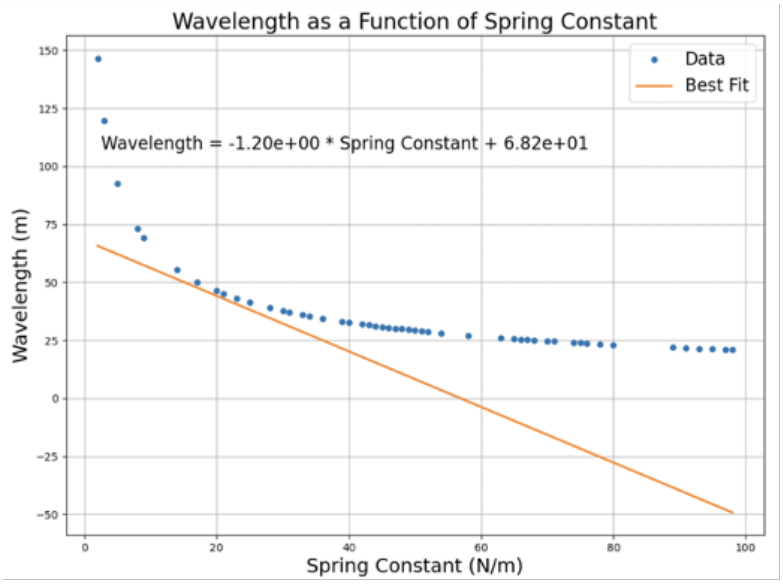


Figure 2: Wavelength as a Function of Spring Constant; Initial Simulation Model with Arbitrary Spring Constant Values

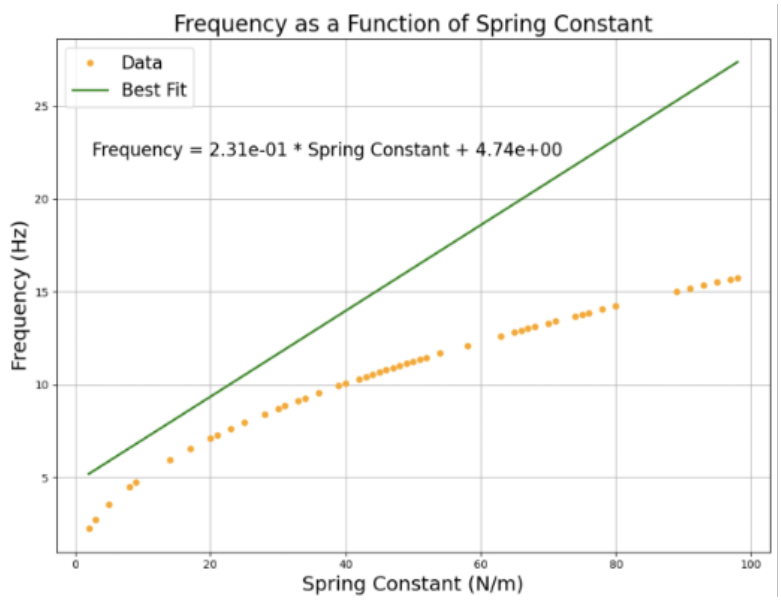


Figure 3: Frequency as a Function of Spring Constant; Initial Simulation Model Using Arbitrary Spring Constant Values

4.2 Refined Model: Realistic Ranges from External Sources

We refined our initial simulation model by implementing empirical data ranges from researched and tested studies. Our refined model's range matched the researched graph's range of the speed of longitudinal waves and tension exerted on the string (4000 to 5200 m/s and 660 and 800 N, respectively). By setting arbitrary values for the frequency and displacement of the string (500 Hz and 0.01m, respectively), 30 values for the wavelengths and spring constants were generated. Figure 4 below depicts our refined model; the results indicate that as the spring constant of a piano string increases, the wavelength of its produced soundwave decreases. The graph's negative slope of the best-fit line demonstrates that the wavelength of a piano string's sound wave is negatively correlated to the string's spring constant.

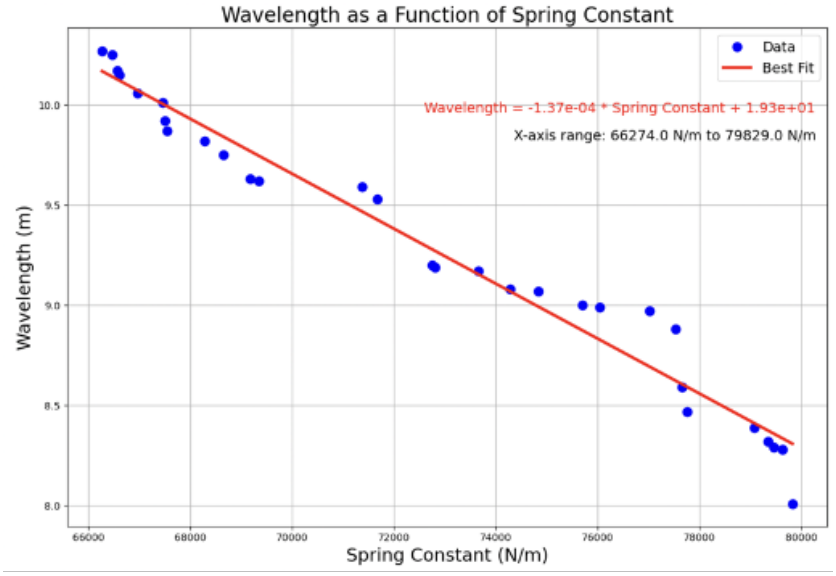


Figure 4: Wavelength as a Function of Spring Constant; Refined Simulation Model Using Range of External Data

4.3 Final Model: Model Data vs. Researched Data

In our final model, each data point from the researched graph was extracted into our model. This allowed our model to compare its generated theoretical

data and the raw values from the chosen graph. The theoretical dataset was generated using the same method as our initial model, which implemented our derived physics equation to determine the sound waves' wavelengths. However, in this final model, the ranges of spring constants were taken from the ranges achieved in model 2, allowing for a more accurate piano string representation. As shown in Figure 5 below, the best-fit slope of the model data indicates that the piano string's soundwave wavelength is negatively correlated with its stiffness. The slope of the raw data best-fit line also states this indirect relationship; however, this slope is relatively smaller in magnitude and appears much weaker than that of the model data.

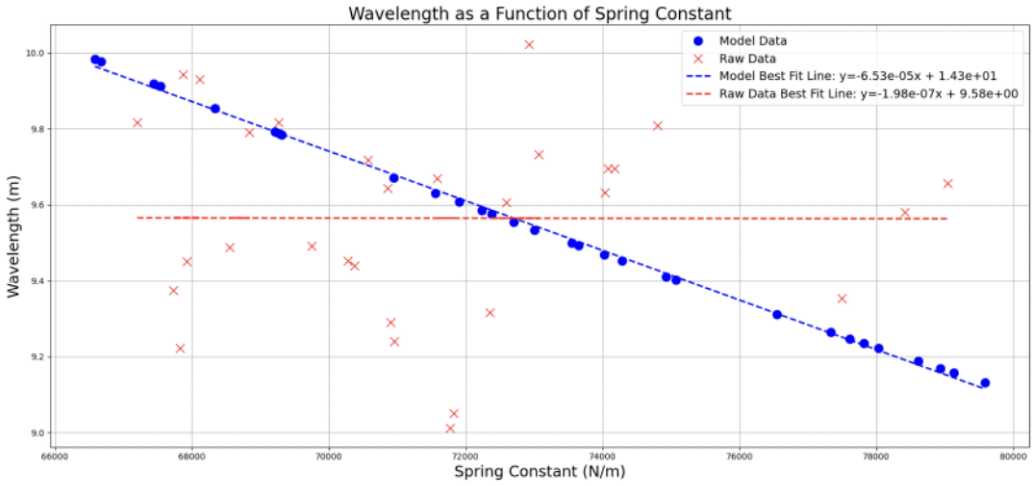


Figure 5: Wavelength as a Function of Spring Constant; Final Refined Simulation Model; Comparison of Theoretical Model and Raw Researched Data

5 Discussion

5.1 Key Findings

This experiment's results confirm an inverse relationship between the spring coefficient of a piano string and the wavelength of the piano string's soundwave. From a purely theoretical perspective, the data from Model 1 confirms

this trend and validates the derived equation:

$$\lambda = \sqrt{\frac{4\rho\pi^3 r^2 l v^2}{k}}$$

The results of model 1 confirm that there is indeed a negative correlation between wavelength and spring constant and a positive correlation between the frequency and the spring constant. In addition, this trend between the wavelength and the piano string's stiffness was still seen in the results of Model 2. Even after incorporating empirical data ranges and information from tested and researched sources, this model displayed the indirect relationship between these variables.

In the final model, there was a slight discrepancy between the theoretical model data and the raw data from the researched graph. As shown in Figure 5, the two slopes of each model's best-fit line remained negative, indicating the negative relationship between wavelength and soundwave. However, the raw data model's slope was far weaker than the theoretical data model, as its data points were more scattered. This reveals the possible inconsistencies in real tested data compared to a theoretical model. From environmental aspects to basic measurement errors, it is difficult to account for all potential things that could impact the research study's results. Specifically, this discrepancy could be due to factors such as variations in the material composition of the string and the impact of the wearing and tearing of the string over time. As these factors were not accounted for, the raw data model appeared scattered and inconsistent relative to the theoretical model.

Upon closer examination of the raw data model, it is evident that there is a negative trend between the wavelength and spring constant between the string stiffness values 66000 and 72000 N/m. Assuming that this data was collected using a standard piano, the inconsistencies in the second portion (after a spring constant value of 72000 N/m) could be due to the more frequent usage of higher keynotes. The assumption that higher notes on a piano are more frequently used could be the reason for the inconsistencies within the raw model, as there would be an increased amount of wear and tear for higher-frequency keys. Despite this inconsistency, the slope of the best-fit line of our raw data model demonstrates a negative correlation between a piano string's wavelength and its stiffness.

Ultimately, this experiment's results conclude that a piano string's stiffness or spring coefficient plays a significant role in shaping the sound it

produces. Our results indicate that a higher spring constant leads to shorter wavelengths and higher frequencies.

5.2 Implications & Significance

The findings of this study significantly contribute to our understanding of wave analysis and string properties within a piano. Our simulation models confirm the inverse relationship between a piano string's stiffness and the wavelength of its produced soundwave. Thus, this research supports those who delve into string tuning and other sections of piano engineering, as it provides a theoretical basis for string tuning and details the significant influence of a piano string's stiffness on its overall sound. By demonstrating how the intrinsic properties of piano strings can impact its sound, this paper can improve the current approach to manufacturing and maintaining pianos.

5.3 Objectives & Future Recommendations

The objectives of this research project were met, as we provided insight into how a piano string's spring coefficient impacts its pitch or sound. This data is mainly helpful to those who work in piano tuning, as it exemplifies the significance of understanding the role of string stiffness in tuning. Our simulation model can serve as a foundational basis for understanding the impact of acoustic properties on a piano string's sound production. To further refine our model, future research could investigate the effects of environmental conditions on our model's raw data. Furthermore, our study could benefit from future studies examining how the string's material composition could impact its production of sound. Through this information, our model would provide a better understanding of the intricate sound waves in our simulation model.

5.4 Limitations

This research study acknowledges several factors that can affect the results and conclusions of our experiment. Firstly, this study relied on a programmed simulation model, causing all our data points to be computer-generated without physical data. Our study lacked our own natural and tested evidence from a physical laboratory. This diminished our ability to collect data, as we had to rely on external sources and online repositories to replace the variables within our derived equations. The lack of use of our physically tested

data and the inability to implement external data sources that may not fully match our research objectives directly affect the accuracy of our experimental results.

From a computational standpoint, the physical resources used for this research were limited to a personal laptop, particularly the MacBook M1. Thus, the lack of computational power could reduce the speed of algorithms and our model, reducing its complexity. This could cause a significant decrease in the precision and accuracy of our data. Implementing higher-power handling devices allows us to benefit from and increase the computational power of our models.

Reflecting on the theoretical framework we derived, the physics we employed was relatively simplistic. Our derivation primarily implemented basic physics principles such as Hooke's and Newton's Second Law of Motion. It did not account for the various factors that could affect our model's results. Our simulation model did not factor in a piano string's environmental components or material imperfections that can change the results. For instance, inconsistencies within the dimensions of the piano string can directly impact the vibrations produced. An uneven string can create deviations in its expected wave behavior [14], ultimately compromising the accuracy of this experiment.

Hence, although our model provides a vital insight into the relationship between string stiffness and wavelength, it only partially considers the many complexities in a real-world scenario.

5.5 Closing Thought

This research serves as a significant contribution to both the fields of theoretical physics and music. By diving into the acoustics of a piano string and determining the relationship between its stiffness and its resulting sound wave wavelength, this study not only advances our current understanding of a piano string's sound waves but invites future researchers to explore the topic further. Specifically, this study invites researchers to dive deeper into the mechanics behind the strings' properties and how this knowledge can be applied in the real world. This information will improve our simulation model and enhance our current understanding of the physics behind the production of sound. Ultimately, this study provides a more thorough perspective of the physics of sound, which enriches our knowledge of physics and sparks the art of music.

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